5.5 Solving Equations by Graphing

Name _____ Period ____ Date _____

Algebra I 5.5 · Student sheet

I can…

- solve linear equations by graphing.
- solve absolute value equations by graphing.
- use graphs to solve real-life problems.

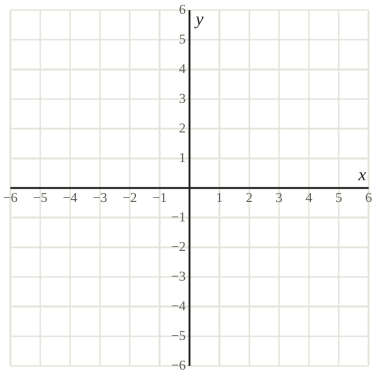
Vocabulary · fill in the missing word

Solving by graphing — to _____ f(x) = g(x), graph y = f(x) and y = g(x); the x-coordinates of the intersection points are the solutions

Part 1 · Two sides, two graphs

Solve −x + 1 = 2x − 5 algebraically: _____

Now graph y = −x + 1 and y = 2x − 5. Intersection: _____



Graph y = |x − 1| and y = 2 on the same grid. Solutions of |x − 1| = 2: _____

Part 2 · Notes: fill in the blanks

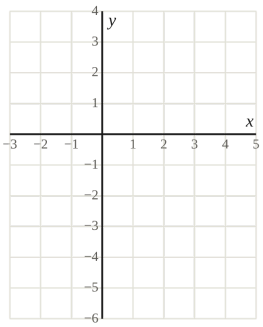
To solve f(x) = g(x), graph y = _____ and y = _____.

The solution is the _____-coordinate of each intersection.

Part 3 · Examples (work them with the class)

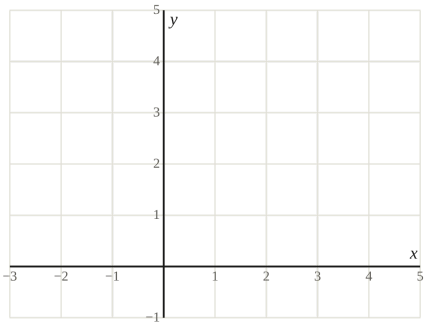
Example 1

Solve $-x + 1 = 2x - 5$ by graphing.



Example 2

Solve $|x - 1| = 2$ by graphing.



Example 3

You have $40 and save $5 a week. Your friend has $10 and saves $15 a week. Graph to find when you have the same amount.

Part 4 · Partner practice

Partner 1

Solve $\frac{1}{2}x + 2 = -x + 5$ by graphing.

Partner 2

Solve $|x + 2| = x + 4$ by graphing.